

Prateeth Rao

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Website: prateeth8.github.io Github: github.com/Prateeth8

VSLAM: github.com/Prateeth8/VSLAM_experiment_web



About

Master's student in Data Science at IIIT-Bengaluru with interests in Visual SLAM, 3D Reconstruction, Scene Graphs, Deepfake Analysis and Remote Sensing. Research aims at enhancing monocular VO using geometric priors and graph neural networks. Actively involved in robotics, GIS analytics, and multimodal perception.

Education

- **M.S. in Data Science**, IIIT-B GPA: 3.75/4.0 2023–2026
- **B.Tech in ECE**, PES University GPA: 8.5/10.0 2019–2023

Experience

Project Intern, COMET Lab, IIIT-B Feb 2023 – Jul 2023

- Developed 5G NR transmitter components and CSI-based DL models outperforming traditional pipelines.
- Published work at IEEE GCITC 2023 under Dr. Satish Kumar.

Senior Intern, MedInn TechLab, PESU Sept 2022 – May 2023

- Designed video analytics pipeline for patient identification in hospitals.
- Mentored junior researchers; published 6 medical imaging papers.

Core Team IoT, C.O.D.S., PES University Oct 2020 – Jan 2022

- Developed ADAS prototypes using sensor fusion and IoT cloud.

Skills

Programming: Python, Embedded C, LaTeX

Tools: NumPy, OpenCV, SciPy, TensorFlow, PyTorch, Kornia, Git, ROS2, GEE, GRASS GIS

Projects

- **Epipolar Graph Geometry for Camera Pose Regression (Thesis,IJCV)**
 - Estimation of Relative Camera Pose by Approximation to 8-point method using Novel graph passing and aggregation modules supervised by Epipolar constraints.

- GitHub Page: Epipolar Graph VO (webpage)
- **Epipolar Diffusion Matcher**
 - LoFTR Matching + Diffusion - Epipolar Diffusion Matcher, Treating noise in images as time varying and supervising with Epipolar Constraints.
- **ARuCo Scene Graph Generation**
 - Instance Segmentation (SAM3) combined with Aruco Detection to form Global Spatial Scene graphs for Visual Localization.
 - Github Link: ARuCo-Scene-Graph
- **Autonomous Agent Global Path planning in self Defined Constrained environment**
 - Self Defined Environment + Initial-goal coordinates + obstacles - Global Path Planning (Dijkstra, A*, D*, RRT etc).
 - Github Link: Global Path Planning
- **Accurate Road with Estimation from using GEE and DL**
 - Real time Road Width Estimation from LandSAT data from GEE of city roads/highways with help of DL modules (segmentation + Pixel width Analysis)
 - Github Link: Road Width Estn
- **6G Channel Diffusion Learning for improved user vector decoding**
 - Modeling Attenuation/ Signal Interference as a time varying diffusion process over the channel.
 - Github Link: 6G Diffusion

Achievements

- CMR scholarship in B.Tech (Last Semester)
- MRD Scholarship in B.Tech (3 semesters)
- Best Paper Award — IEEE CENTCON 2023

Publications

Google Scholar link - https://scholar.google.com/citations?user=Vk1L-_OAAAAJ&hl=en

Volunteer Work

- Volunteer + Poster presenter — RISE 2025 (IIIT-B)
- TPC — IEEE CONECCT 2024; Volunteer — ICVGIP-24
- Organizer/mentor — IoT Hackathon and Workshop (PES University)

Interests

Yoga, gardening, puzzles, music (Classical, Bollywood, EDM) and learning historical findings.